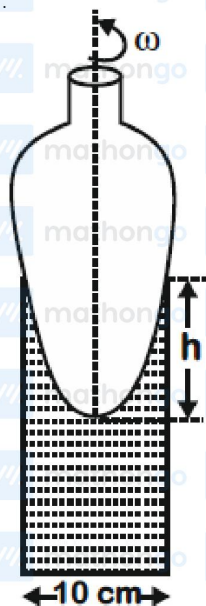


Q1 JEE Main 2020 - 2 September (Morning)

A cylindrical vessel containing a liquid is rotated about its axis so that the liquid rises at its sides as shown in the figure. The radius of vessel is 5cm and the angular speed of rotation is $\omega \text{ rad s}^{-1}$. The difference in the height, h (in cm) of liquid at the centre of vessel and at the side will be



- (A) $\frac{2\omega^2}{25g}$
- (B) $\frac{5\omega^2}{2g}$
- (C) $\frac{2\omega^2}{5g}$
- (D) $\frac{25\omega^2}{2g}$

Q2 JEE Main 2020 - 2 September (Evening)

A capillary tube made of glass of radius 0.15 mm is dipped vertically in a beaker filled with methylene iodide (surface tension = 0.05 Nm^{-1} , density = 667 kgm^{-3}) which rises to height h in the tube. It is observed that the two tangents drawn from liquid-glass interfaces (from opp. sides of the capillary) make an angle of 60° with one another. Then h is close to ($g = 10 \text{ ms}^{-2}$)

- (A) 0.049m
- (B) 0.087m

Fluid Mechanics

JEE Main 2020 Chapterwise

Questions with Answer Keys

Physics

(C) 0.137m

(D) 0.172m

Q3 JEE Main 2020 - 3 September (Morning)

Pressure inside two soap bubbles are 1.01 and 1.02 atmosphere, respectively. The ratio of their volumes is

(A) 02:01

(B) 0.8: 1

(C) 04:01

(D) 08:01

Q4 JEE Main 2020 - 3 September (Morning)

When a long glass capillary tube of radius 0.015cm is dipped in a liquid, the liquid rises to a height of 15cm within it. If the contact angle between the liquid and glass is close to 0° , the surface tension of the liquid, in milliNewton m^{-1} , is $[\rho_{(\text{liquid})} = 900\text{kgm}^{-3}, g = 10\text{ms}^{-2}]$ (Give answer in closest integer)_____.

Q5 JEE Main 2020 - 4 September (Evening)

Two identical cylindrical vessels are kept on the ground and each contain the same liquid of density d . The area of the base of both vessels is S but the height of liquid in one vessel is x_1 and in the other, x_2 . When both cylinders are connected through a pipe of negligible volume very close to the bottom, the liquid flows from one vessel to the other until it comes to equilibrium at a new height. The change in energy of the system in the process is

(A) $gdS(x_2 + x_1)^2$

(B) $\frac{1}{4}gdS(x_2 - x_1)^2$

(C) $\frac{3}{4}gdS(x_2 - x_1)^2$

(D) $gdS(x_2^2 + x_1^2)$

Q6 JEE Main 2020 - 5 September (Morning)

A hollow spherical shell at outer radius R floats just submerged under the water surface. The inner radius of the shell is r . If the specific gravity of the shell material is $\frac{27}{8}$ w.r.t water, the value of r is

- (A) $\frac{2}{3}R$
- (B) $\frac{4}{9}R$
- (C) $\frac{1}{3}R$
- (D) $\frac{8}{9}R$

Q7 JEE Main 2020 - 6 September (Evening)

A fluid is flowing through a horizontal pipe of varying cross-section, with speed $v \text{ ms}^{-1}$ at a point where the pressure is P pascal. At another point where pressure is $\frac{P}{2}$ pascal its speed is $V \text{ ms}^{-1}$. If the density of the fluid is $\rho \text{ kgm}^{-3}$ and the flow is streamline, then V is equal to :

- (A) $\sqrt{\frac{P}{\rho} + v^2}$
- (B) $\sqrt{\frac{2P}{\rho} + v^2}$
- (C) $\sqrt{\frac{P}{\rho} + v}$
- (D) $\sqrt{\frac{P}{2\rho} + v^2}$

Q8 JEE Main 2020 - 7 January (Evening)

An ideal fluid flows (laminar flow) through a pipe of non-uniform diameter. The maximum and minimum diameters of the pipes are 6.4 cm and 4.8 cm , respectively. The ratio of the minimum and the maximum velocities of fluid in this pipe is :

- (A) $\frac{81}{256}$
- (B) $\frac{9}{16}$
- (C) $\frac{3}{4}$
- (D) $\frac{3}{16}$

Q9 JEE Main 2020 - 8 January (Morning)

A leak proof cylinder of length 1 m , made of a metal which has very low coefficient of expansion is floating vertically in water at 0°C such that its height above the water surface is 20 cm . When the

temperature of water is increased to 4°C , the height of the cylinder above the water surface becomes 21cm . The density of water at $T = 4^{\circ}\text{C}$, relative to the density at $T = 0^{\circ}\text{C}$ is close to:

- (A) 1.01
- (B) 1.03
- (C) 2.01
- (D) 1.04

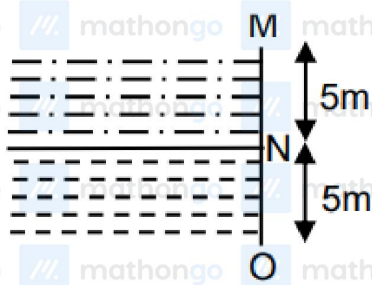
Q10 JEE Main 2020 - 8 January (Morning)

A solid sphere of density $\rho = \rho_0 \left(1 - \frac{r^2}{R^2}\right)$, $0 < r \leq R$ just floats in a liquid. The density of liquid is (r is distance from centre of sphere)

- (A) $\frac{2}{5}\rho_0$
- (B) $\frac{5}{2}\rho_0$
- (C) $\frac{3}{5}\rho_0$
- (D) ρ_0

Q11 JEE Main 2020 - 8 January (Evening)

Two liquid columns of same height 5 m and densities ρ and 2ρ are filled in a container of uniform cross sectional area. Then ratio of force exerted by the liquid on upper half of the wall to lower half of

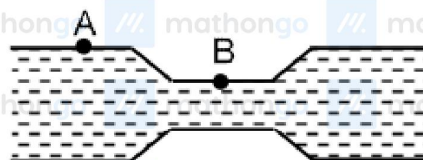


the wall is.

- (A) $\frac{1}{4}$
- (B) $\frac{1}{2}$
- (C) $\frac{1}{3}$
- (D) $\frac{2}{3}$

Q12 JEE Main 2020 - 9 January (Morning)

Water flows in a horizontal tube (see figure). The pressure of water changes by 700 Nm^{-2} between A and B where the area of cross section are 40 cm^2 and 20 cm^2 , respectively. Find the rate of flow of



water through the tube. (density of water = 1000 kgm^{-3})

- (A) $2732 \text{ cm}^3/\text{s}$
- (B) $2142 \text{ cm}^3/\text{s}$
- (C) $1832 \text{ cm}^3/\text{s}$
- (D) $3218 \text{ cm}^3/\text{s}$

Q13 JEE Main 2020 - 4 September (Morning)

A air bubble of radius 1 cm in water has an upward acceleration 9.8 cm/s^2 . The density of water is 1 gm/cm^3 and water offers negligible drag force on the bubble. The mass of the bubble is ($g = 980 \text{ cm/s}^2$)

- (A) 1.52 gm
- (B) 4.51 gm
- (C) 3.15 gm
- (D) 4.15 gm

Fluid Mechanics

Questions with Answer Keys

JEE Main 2020 Chapterwise

Physics

Answer Key

Q1 (D)

Q2 (B)

Q3 (D)

Q4 (101)

Q5 (B)

Q6 (D)

Q7 (A)

Q8 (B)

Q9 (A)

Q10 (A)

Q11 (A)

Q12 (A)

Q13 (D)

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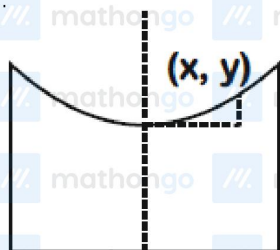
Fluid Mechanics

Hints & Solutions

JEE Main 2020 Chapterwise

Physics

Q1



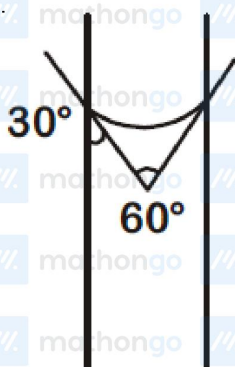
$$y = \frac{\omega^2 x^2}{2g}$$

$$x = R$$

$$\Rightarrow y = \frac{25\omega^2}{2g}$$

Q2

Angle of contact = 30°



$$h = \frac{2T \cos \theta}{r \rho g} = \frac{2 \times 0.05 \times \left(\frac{\sqrt{3}}{2}\right)}{0.15 \times 10^{-3} \times 667 \times 10} = 0.087 \text{ m}$$

Q3

$$P_1 = P_0 + \frac{4T}{r_1}$$

$$P_2 = P_0 + \frac{4T}{r_2}$$

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Fluid Mechanics

Hints & Solutions

JEE Main 2020 Chapterwise

Physics

$$\frac{r_1}{r_2} = 2$$

$$\Rightarrow \frac{v_1}{v_2} = 8$$

Q4

$$h = \frac{2T \cos \theta}{\rho g r}$$

$$\Rightarrow T = \frac{\rho g r \times h}{2 \cos \theta} = \frac{900 \times 10 \times 15 \times 15}{100 \times 1000 \times 100 \times 2}$$

$$\Rightarrow T = 0.10125 \text{ Nm}^{-1}$$

$$\Rightarrow T = 101.25 \approx 101 \text{ milliNewtonm}^{-1}$$

Q5

$$|\text{Change in energy}| = U_i - U_f$$

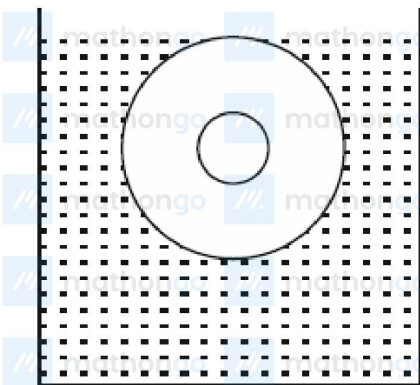
$$= (dsx_1) g \frac{x_1}{2} + (dsx_2) g \frac{x_2}{2}$$

$$= -2 \times \left\{ ds \left(\frac{x_1 + x_2}{2} \right) \right\} g \left(\frac{x_1 + x_2}{4} \right)$$

$$= \frac{1}{4} g ds (x_2 - x_1)^2$$

Q6

$$\frac{4}{3} \pi R^3 \times \rho_w \times g = \frac{4}{3} \pi (R^3 - r^3) \rho_m \times g$$



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Fluid Mechanics

Hints & Solutions

JEE Main 2020 Chapterwise

Physics

$$\Rightarrow R^3 = (R^3 - r^3) \times \frac{27}{8}$$

$$\Rightarrow r = \left(\frac{19}{27}\right)^{\frac{1}{3}} R \approx \frac{8}{9} R$$

Q7

$$P + \frac{1}{2} \rho v^2 = \frac{P}{2} + \frac{1}{2} \rho V^2$$

$$\Rightarrow V = \sqrt{\frac{P}{\rho} + v^2}$$

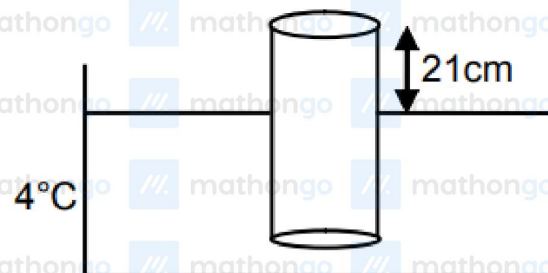
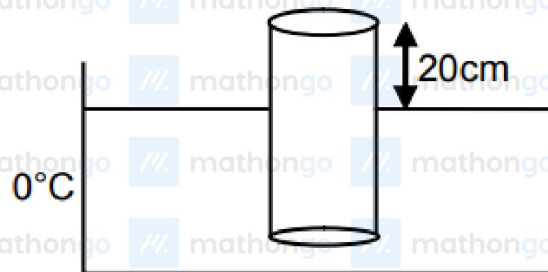
Q8

Using equation of continuity

$$A_1 V_1 = A_2 V_2$$

$$\frac{V_1}{V_2} = \frac{A_2}{A_1} = \left(\frac{4.8}{6.4}\right)^2 = \frac{9}{16}$$

Q9



$$mg = A (80) \rho_{0^\circ C} g$$

$$mg = A (79) \rho_{4^\circ C} g$$

$$\frac{\rho_{4^\circ C}}{\rho_{0^\circ C}} = \frac{80}{79} = 1.01$$

Q10

$$\rho = \rho_0 \left(1 - \frac{r^2}{R^2} \right)$$

$$mg = B$$

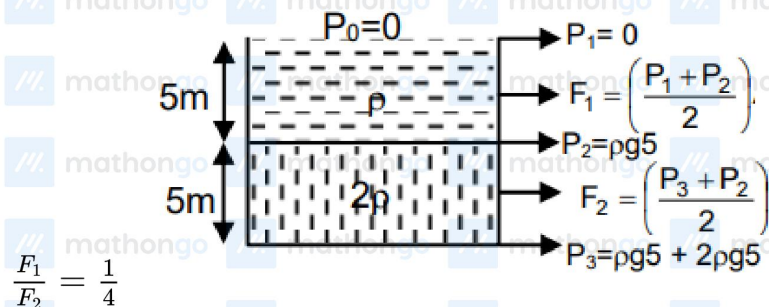
$$\int_0^R \rho (4\pi r^2 dr) = \rho_L \frac{4}{3} \pi R^3$$

$$\int_0^R \rho_0 \left(1 - \frac{r^2}{R^2} \right) 4\pi r^2 dr = \rho_L \frac{4}{3} \pi R^3$$

$$\int_0^R \rho_0 4\pi \left(r^2 - \frac{r^4}{R^2} \right) dr = \rho_0 4\pi \left(\frac{r^3}{3} - \frac{r^5}{5R^2} \right) \Big|_0^R = \rho_L \frac{4}{3} \pi R^3$$

$$\frac{2}{5} \rho_0 = \rho_L$$

Q11



Q12

using equation of continuity

$$40 V_A = 20 V_B$$

$$\Rightarrow 2V_A = V_B$$

Using Bernoulli's equation

$$P_A + \frac{1}{2} \rho V_A^2 = P_B + \frac{1}{2} \rho V_B^2$$

$$\Rightarrow P_A - P_B = \frac{1}{2} \rho (V_B^2 - V_A^2)$$

$$\Rightarrow \Delta P = \frac{1}{2} 1000 \left(V_B^2 - \frac{V_B^2}{4} \right)$$

$$\Rightarrow \Delta P = 500 \times \frac{3V_B^2}{4}$$

$$\Rightarrow V_B = \sqrt{\frac{(\Delta P) \times 4}{1500}} = \sqrt{\frac{(700) \times 4}{1500}} \text{ m/s}$$

$$\text{Volume flow rate} = 20 \times 100 \times V_B = 2732 \text{ cm}^3/\text{s}$$

Q13

$$\frac{4}{3} \pi r^3 \rho_w g - mg = ma$$

$$\Rightarrow m = \frac{\frac{4}{3} \pi r^3 \rho_w g}{g+a}$$

$$= 4.15 \text{ gm}$$